Panshul Jindal

ai23btech11018@iith.ac.in | panshul.jindal7@gmail.com | +91 7347446449 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.14
Dr Kitchlu Public Senior Secondary School, Moga	Class XII (CBSE)	2023	94.40%
Sacred Heart School, Dosanj Road, Moga	Class X (ICSE)	2021	96.80%

EXPERIENCE

Schema-Agnostic Generic Embeddings for Knowledge Graphs

Software Engineering Intern @ Google

- Unified multiple graph ML pipelines for recommendation tasks into a single self-supervised framework, eliminating ~2 FTE of engineering effort per independent pipeline.
- Improved performance across downstream tasks by up to **100%**, while reducing overall training time versus multiple independent pipelines.
- Designed and Deployed an **easily extendable** generic embedding framework for future development following best engineering practice.

SKILLS

Programming Languages: Python, C, C++, SQL, RISC-V (Assembly), Golang

Development: React, Node.js, Express, Next.js

PROJECTS

Equity Research & Quantitative Risk-Return Analysis | EquityX -- 2nd Place

- **Selected and evaluated Saksoft Ltd. and Trident Ltd.** as small-cap and mid-cap investment candidates, respectively, using historical price trends, financial performance, valuation multiples, leverage and peer benchmarks to assess **risk-return potential**.
- **Analyzed multi-year financial trends**, including revenue, net profit, profit margins, debt-to-equity, P/E and EV/EBITDA; identified **2.13× net-profit growth for Saksoft** and **2.2× for Trident** over the analyzed period.
- **Developed investment theses with explicit downside protection**, setting target/stop-loss levels of **375/315 for Saksoft** and **70/33 for Trident**, and presented the analysis in an intra-college equity research competition, finishing **2nd among participants**.

RISC-V Assembler & Cache Simulator | C++

- Developed a RISC-V 64 bit assembler (CLI) from scratch with full support for all instruction formats (R, I, S, B, U, J)
- Built a configurable RISC-V cache simulator supporting direct-mapped, fully associative, and set-associative caching with FIFO and LRU replacement policies. Validated using Ripes & RISC-V tools

IIT Hyderabad Gaming Club, Glitch, Website | React, Next.js, Golang

- Engineered and launched the official web platform for Glitch, successfully hosting the Inter-Hostel Championship 2025

IIT Hyderabad Library Group Discussion Rooms Booking Software | React, Next.js, Golang

- Engineered end-to-end frontend and backend integrations to handle concurrent user bookings and live schedule updates

Implementation of RAFT and ZooKeeper Atomic Broadcast Algorithms from Scratch | Golang

- Designed an in-memory network simulator to validate protocol behavior and leader election performance

RELEVANT COURSES

Artificial Intelligence: Speech Systems, Data Mining, Advance Topics in Machine Learning, Distributed AI Training in Large Scale GPU Clusters, Topics in Deep Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Deep Learning, Statistical Learning Theory, Online Learning, Foundations of Machine Learning, Programming for AI, Introduction to Modern AI

Computer Science: Distributed Systems, Computer Networks, Computer Architecture, Database Management Systems, Data Structures and Applications, Algorithms, OS-1, Compilers-1, Introduction to Programming

Mathematics: Probability and Random Variables, Applied Statistics, Matrix Theory, Convex Optimization, Advanced Optimization, Concentration Inequalities, Differential Equations, Series of Functions, Calculus

Others: Control Systems, Linear Systems and Signal Processing

EXTRACURRICULAR

IIT Hyderabad Representative: 40th Inter-IIT Aquatics Meet (2026), 38th Inter-IIT Aquatics Meet (2024), 56th Inter-IIT Sports Meet - Chess (2023)

State Representative, Punjab: SGFI U-14 Chess Nationals (2017), National Senior Open Chess Championship (2018), SGFI U-17 Chess Nationals (2019)

Northern Region Representative, CISCE: U-17 Swimming National (2018), U-17 Chess National (2018)

Second Runner Up Scribble Pad IIT-H JavaScript Hackathon 2024

POSITIONS OF RESPONSIBILITY

Senior Core, Machine Learning Domain - Tinkerers' Lab, IIT Hyderabad

Senior Core, Epoch - AI/ML Club, IIT Hyderabad

Problem Statement Lead, Inter-IIT Tech Meet 2025:

Lead the college in Conversational AI challenge by Observe.AI with challenge of developing a chabot to answer queries on call center data causally and with full interpretability and producing a benchmark system to evaluate it's quality.

Dev Core, Lambda - Software Development Club, IIT Hyderabad

Class Representative, B.Tech Artificial Intelligence Batch, Class of 2027